Ferroelectric Vortex GPU Platform

Complete Engineering Specification

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This document provides the complete engineering specification for a graphics processor that uses a fixed neural network (neural shader) implemented on a crossbar array of ferroelectric cells to compute pixel colors. All cell geometries, material compositions, fabrication steps, circuit designs, operating procedures, qualification tests, and acceptance criteria are provided explicitly. Every parameter required to build, test, and operate the baseline device is included. The design is released as a public, patent-free engineering specification. An advanced optimization configuration is offered separately and is described at the end of this document.

This is a 6DUFT inspired technology.

Abstract

This document is the complete engineering specification for a graphics processor built on ferroelectric vortex memory cells, using a fixed neural network as the pixel shader.

Instead of programmable shader cores that execute sequential instructions, the device computes pixel colors using a fixed neural network implemented as a series of matrix-vector multiplications on a crossbar array. The neural network has 10 layers of 1024×1024 matrices. Each layer is a single parallel operation.

The baseline device provides:

- **Internal pixel rate:** 196 Gp/s
- **External pixel rate:** 176.4 Gp/s
- **MAC throughput:** 2.06×10^{18} MAC/s
- **Power:** 560 mW
- **Efficiency:** 350 Gp/J
- **Die area:** 71 mm^2
- **Cost:** $9.11

The device uses standard 28 nm CMOS with hafnium zirconium oxide ($Hf_{0.5}Zr_{0.5}O_2$, HZO) atomic layer deposition. No exotic materials are required. The complete fabrication flow has 18 steps and 10 masks.

The document covers architecture, cell geometry and materials, fabrication flow, circuit design, operating principles, performance specifications, verification and qualification, failure modes and mitigations, packaging, and an advanced optimization configuration available separately.

The design is released openly. No patent is sought.

This is a 6DUFT inspired technology.

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Part I
Introduction and Overview

1 Purpose and Scope

This document is the complete engineering specification for a graphics processor built on ferroelectric vortex memory cells, using a fixed neural network as the pixel shader.

It provides every parameter, process step, circuit specification, and procedure required to build, test, and operate the baseline device.

The document is written for:

- Foundry process engineers who will fabricate the device.
- Circuit designers who will integrate the device.
- Test engineers who will qualify the device.
- System engineers who will deploy the device.

No prior knowledge of any theoretical framework is required. All necessary engineering information is contained in this document.

2 What the Device Is

The device is a graphics processor that computes pixel colors using a neural network implemented as a series of matrix-vector multiplications.

Each neural shader layer is a 1024×1024 crossbar of ferroelectric cells. The crossbar stores the neural network weights. Input pixel attributes are applied as voltages on dual word lines. Output pixel values are read from the bit lines.

The neural network has 10 layers. Each layer is a single parallel operation. The output of each layer passes through an on-chip analog rectifier (the activation function) before entering the next layer.

The device accepts compressed scene data over a PCIe 5.0 interface and outputs compressed frames over the same interface. On-chip decompression and encoding reduce the external data bandwidth by a fixed factor.

3 Why a Neural Shader

Traditional GPUs use programmable shader cores that execute sequential instructions for each pixel. This requires instruction fetch and decode, register file access, arithmetic operations, memory access, and control flow. Each of these steps consumes power and takes time.

A neural shader replaces the sequential shader program with a fixed neural network. The neural network is a series of matrix-vector multiplications. The crossbar performs each multiplication in a single parallel operation.

The result is a pixel shading pipeline reduced to 10 matrix-vector multiplications, each taking picoseconds.

4 What the Device Is Not

The device is not:

- A traditional GPU. It does not execute programmable shader code.
- A ray tracing accelerator. It cannot perform ray tracing.
- A general-purpose compute device. It is specialized for neural network-based graphics rendering.
- Field-programmable. The neural network weights are fixed at fabrication.
- A training device. The neural network is trained offline and the weights are programmed at fabrication.

5 Key Advantages

- **External pixel rate:** 176.4 Gp/s, which is 70.6× a typical flagship GPU's display output.
- **Very low power:** 560 mW total, which is 804× lower than a typical flagship GPU.
- **High efficiency:** 350 Gp/J, which is 350× a typical flagship GPU.
- **Small die area:** 71 mm^2, which is 8.6× smaller than a typical flagship GPU.
- **Low cost:** $9.11 per chip.
- **Non-volatile weights:** No memory initialization required. Instant boot.
- **CMOS compatible:** Standard 28 nm CMOS with HZO atomic layer deposition.
- **No exotic materials.**

6 Key Limitations

- **Baseline internal pixel rate:** 196 Gp/s, which is 0.44× a flagship GPU.
- **No programmability:** The neural network weights are fixed at fabrication.
- **No ray tracing:** The neural shader cannot perform ray tracing.
- **Low weight precision:** 1-bit weights at the baseline configuration.

7 Device Positioning

The baseline device is positioned for:

- Embedded graphics
- Low-power desktop and mobile systems
- Applications where the neural shader has been trained for the target rendering style
- Multi-display systems where the high external pixel rate is beneficial

An advanced optimization configuration with higher internal pixel rate, lower power, and higher efficiency is available separately and is described at the end of this document.

Part II
Device Architecture

8 Neural Network Structure

The neural shader is a 10-layer fully-connected neural network with 1024 inputs and 1024 outputs per layer.

Layer	Size	Function
Input	64 attributes	Pixel attributes (differential)
1	1024×1024	Feature extraction
2	1024×1024	Feature processing
3	1024×1024	Feature processing
4	1024×1024	Feature processing
5	1024×1024	Feature processing
6	1024×1024	Feature processing
7	1024×1024	Feature processing
8	1024×1024	Feature processing
9	1024×1024	Feature processing
10	1024×1024	Output

Each layer is a matrix-vector multiplication. The input to each layer is the output of the previous layer after the on-chip analog rectifier. The output layer produces 1024 values, of which 4 are used for RGBA color output.

The weights are fixed at fabrication and are supplied to the foundry as a mask pattern. See Section "Weight Data Interface" for the exact data format.

9 Dual Word Line Architecture

The signed weights are represented using dual word lines:

- **WL^+:** Carries the positive input $+x_i$.
- **WL^-:** Carries the negative input $-x_i$.

Cells with a positive weight connect to WL^+. Cells with a negative weight connect to WL^-. The current on bit line j is the sum over all inputs, giving the correct signed matrix-vector product.

10 On-Chip Analog Rectifier

The activation function is an on-chip analog rectifier (ReLU).

Parameter	Value
Type	Current-mirror rectifier + buffer chain
Input range	0–50 mV
Output range	0–50 mV
Rectification	Blocks negative currents
Supply voltage	1.0 V

The rectifier is placed between each pair of neural network layers.

11 Scene Graph Decompressor

The scene graph decompressor expands compressed scene data into pixel attributes.

Parameter	Value
Input interface	PCIe 5.0, 64 GB/s
Input data width	64 bytes per clock
Input clock	500 MHz
Output data width	256 bytes per pixel block
Output rate	11.3 TB/s peak
Output clock	500 MHz
Gates	1 million
Supply voltage	1.0 V
Power	2 mW

The compression algorithm is not part of this specification. The decompressor block is delivered as a mask-ready RTL netlist.

12 Frame Encoder

The frame encoder compresses the rendered output frames for transmission over the PCIe interface.

Parameter	Value
Input data width	4 bytes per pixel (RGBA)
Input rate	233.9 TB/s peak
Input clock	500 MHz
Output interface	PCIe 5.0, 64 GB/s
Output data width	64 bytes per clock
Output clock	500 MHz
Gates	2 million
Supply voltage	1.0 V
Power	10 mW

The compression algorithm is not part of this specification. The encoder block is delivered as a mask-ready RTL netlist.

13 Internal and External Pixel Rates

13.1 Internal Pixel Rate

The internal pixel rate is the rate at which the device processes pixels through the neural shader.

$$\text{Pixel rate}_{\text{internal}} = \frac{N_{\text{active}}}{t_{\text{compute}}} \tag{1}$$

where N_{active} is the number of simultaneously active neural networks and t_{compute} is the compute time per layer.

At the baseline configuration: $N_{\text{active}} = 14$, $t_{\text{compute}} = 71.4$ ps, giving 196 Gp/s.

13.2 External Pixel Rate

The external pixel rate is the rate at which rendered frames are transferred to the host over the PCIe 5.0 interface.

At the baseline configuration: 176.4 Gp/s.

13.3 MAC Throughput

$$\text{MAC rate} = \text{Pixel rate} \times 10 \times 1024 \times 1024 \tag{2}$$

At the baseline configuration:

$$\text{MAC rate} = 196 \times 10^9 \times 10{,}485{,}760 = 2.06 \times 10^{18} \text{ MAC/s} \tag{3}$$

14 Pipeline Architecture

At any instant, all 10 layers are active, each processing a different pixel. A pixel exits layer 10 after 10 compute cycles.

Parameter	**Value**
Pipeline depth	10 layers
Layer compute time	71.4 ps (baseline)
Pixel exit time	10×71.4 ps $= 714$ ps
Active networks	14 (baseline)

15 Weight Data Interface

The neural network weights are supplied to the foundry as a binary mask pattern file. The file format is as follows:

Field	**Size**	**Description**
Header	64 bytes	Magic number, version, layer count
Layer 1 weights	1024×1024 bits	Row-major order, 1 bit per cell
Layer 2 weights	1024×1024 bits	Row-major order
$\vdots$	$\vdots$	$\vdots$
Layer 10 weights	1024×1024 bits	Row-major order
Trailer	16 bytes	Checksum

A bit value of 1 indicates a conductive cell. A bit value of 0 indicates an insulating cell. The file is used directly to generate mask M5 (disk pattern).

The total weight data size is $10 \times 1024 \times 1024$ bits = 1.31 MB.

16 Clock Domains

The device uses two clock domains:

Domain	Frequency	Used by
PCIe / logic	500 MHz	Decompressor, encoder, control logic
Compute	Layer compute rate (14.0 GHz baseline)	Neural shader crossbar, drivers, comparator

The two clock domains are synchronized through a standard asynchronous FIFO at the boundary between the decompressor output and the neural shader input, and at the boundary between the neural shader output and the encoder input.

Part III
Cell Geometry and Materials

17 Cell Geometry

Parameter	Value
Cell size	120 nm × 120 nm
Disk diameter	80 nm
Pillar diameter	60 nm
Pillar height	1.96 nm
Film thickness	14.83 nm
Annular width	10 nm

18 Annular Area

The conductive region of the cell has an annular area of 452.1 nm^2 at the baseline configuration.

19 Material Properties

19.1 Hafnium Zirconium Oxide

Property	Value
Composition	$Hf_{0.5}Zr_{0.5}O_2$
Film thickness	14.83 nm
Coercive field (25°C)	1.49 MV/cm
Coercive field (85°C)	1.18 MV/cm
Curie temperature	579 K
Remanent polarization	13.07 μC/cm^2
Dielectric constant	30
Resistivity	1.284×10^{-4} Ω·m
Film uniformity	$\pm 1\%$ within wafer

19.2 Conductive Region Properties

Property	Value
Conductive region effective radius	2.31 nm
Conductive region conductivity	7.79×10^3 S/m

19.3 Electrode Materials

Electrode	Material	Thickness
Bottom	TiN	50 nm
Top	TiN	50 nm
Pillar	TiN	1.96 nm
Etch stop	Al_2O_3	2 nm

20 Cell Electrical Characteristics

Parameter	Value
Resistance ($n = 1$)	4,212 Ω
Resistance ($n = 0$)	$> 10^{12}$ Ω
ON/OFF ratio	$> 10^9$
Capacitance	97.8 aF
Settling time	0.412 ps
Read voltage	9.886 mV
Read current ($n = 1$)	2.347 μA

Part IV
Fabrication Flow

21 Overview

The fabrication flow has 18 steps and uses 10 masks. The flow uses standard 28 nm CMOS front-end processing with HZO atomic layer deposition for the ferroelectric layer.

22 Step-by-Step Process Flow

22.1 Step 1: Substrate Preparation

Parameter	Value
Wafer size	300 mm
Substrate	Silicon (100)
Thermal SiO_2	100 nm ± 2 nm
Cleaning	RCA clean before oxide
Surface preparation	HF dip, DI rinse, N_2 dry

22.2 Step 2: FinFET Fabrication

Standard 28 nm CMOS front-end.

Parameter	Value
Gate length	28 nm
Fins	1-fin
Gate oxide	High-κ (HfO_2, EOT 3 nm)
Source/drain	$CoSi_2$
Threshold voltage	0.3 V
On-current (1-fin)	50 μA

22.3 Step 3: Bottom Electrode Deposition

Parameter	Value
Material	TiN
Thickness	50 nm ± 1 nm
Method	ALD
Temperature	300°C
Precursors	$TiCl_4$ + NH_3
Growth rate	0.05 nm/cycle
Cycles	1000

22.4 Step 4: Etch Stop Deposition

Parameter	Value
Material	Al_2O_3
Thickness	2 nm ± 0.2 nm
Method	ALD
Temperature	300°C
Precursors	$Al(CH_3)_3$ + H_2O

22.5 Step 5: Pillar Layer Deposition

Parameter	Value
Material	TiN
Thickness	1.96 nm ± 0.1 nm
Method	ALD
Temperature	300°C

22.6 Step 6: Pillar Patterning

Parameter	Value
Diameter	60 nm
Method	Immersion lithography (193 nm)
Alignment tolerance	±2 nm

22.7 Step 7: Pillar Etch

Parameter	Value
Chemistry	Cl_2/Ar
Method	RIE
Etch rate	∼1 nm/s
Etch stop	Al_2O_3 layer
Overetch	10%
Selectivity to Al_2O_3	>10:1

22.8 Step 8: Etch Stop Removal

Parameter	Value
Etchant	Dilute HF (0.1%)
Al_2O_3 etch rate	0.5 nm/s
TiN etch rate	<0.01 nm/s
Etch time	4 s (2× overetch)

After this step, the Al_2O_3 remains only under the TiN pillar.

22.9 Step 8b: Post-Etch Cleaning

Step	Description
Rinse	Deionized water, 30 s
Dry	N_2 blow dry, 30 s
Plasma clean	Ar plasma, 30 s at 50 W

22.10 Step 9: HZO Deposition

Parameter	Value
Material	$Hf_{0.5}Zr_{0.5}O_2$
Target thickness	14.83 nm
Method	ALD
Deposition temperature	250°C ± 2°C
Chamber pressure	0.5–2 Torr

ALD Recipe:

Step	Precursor	Pulse time	Purge time
1	$HfCl_4$	0.1 s	3 s
2	H_2O	0.1 s	3 s
3	$ZrCl_4$	0.1 s	3 s
4	H_2O	0.1 s	3 s

Cycle ratio: 1 Hf : 1 Zr.
HfO_2 growth rate: 0.08 nm/cycle.
ZrO_2 growth rate: 0.10 nm/cycle.
Supercycle growth: 0.18 nm.
Supercycles required: $14.83/0.18 = 82$.
Actual thickness: $82 \times 0.18 = 14.76$ nm.
Deviation: -0.47% (within $\pm 1\%$ tolerance).
Deposition time: $82 \times 12.4 = 17$ minutes.
Precursor lines: 150–180°C.
Exhaust: corrosion-resistant (HCl byproduct).
Scrubber: wet type for HCl.
Process note: The film must be annealed within 1 hour of deposition to prevent moisture absorption.

22.11 Step 10: RTA Annealing

Parameter	Value
Method	Rapid Thermal Annealing
Temperature	450°C
Duration	20 s
Atmosphere	N_2 (5N purity)
Ramp rate	50°C/s

22.12 Step 11: Top Electrode Deposition

Parameter	Value
Material	TiN
Thickness	50 nm ± 1 nm
Method	ALD
Temperature	300°C

22.13 Step 12: Top Electrode Patterning

Parameter	Value
Pattern	Circle
Diameter	80 nm
Alignment	Centered on pillar
Tolerance	±2 nm

22.14 Step 13: Disk Etch

Parameter	Value
Stack	TiN (50 nm) + HZO (14.83 nm) + TiN (50 nm)
Chemistry (TiN)	Cl_2/Ar
Chemistry (HZO)	BCl_3/Ar
Etch stop	SiO_2 buried oxide
Selectivity to SiO_2	>3:1

Multi-step etch:

1. TiN etch (50 nm): Cl_2/Ar, 50 s.
2. HZO etch (14.83 nm): BCl_3/Ar, 15 s.
3. TiN etch (50 nm): Cl_2/Ar, 50 s.

Endpoint detection: optical emission spectroscopy. Ti line at 363.5 nm, Hf line at 313.5 nm.

22.15 Step 14: Post-Etch Anneal

Parameter	Value
Temperature	400°C
Duration	30 min
Atmosphere	O_2
Purpose	Repair sidewall damage

22.16 Step 15: Passivation

Parameter	Value
Material	SiN (50 nm) + SiO_2 (50 nm)
Method	PECVD
Temperature	350°C
Stress	<200 MPa

22.17 Step 16: Contact Via Formation

Parameter	Value
Via diameter	50 nm
Fill material	Tungsten (CVD)
Barrier	TiN (5 nm)
Resistivity	5.6×10^{-8} Ω·m

22.18 Step 17: Cu Interconnect

Parameter	Value
Thickness	1000 nm
Barrier	Ta/TaN (10 nm)
Seed	Cu (50 nm PVD)
Fill	Cu electroplating
CMP	After fill
Sheet resistance	2.0 mΩ/sq

22.19 Step 18: Final Anneal

Parameter	Value
Temperature	400°C
Duration	30 min
Atmosphere	Forming gas (5% H_2 in N_2)
Purpose	Contact sintering, interface passivation

23 Mask Set

Mask	Purpose	CD	Type
M1	FinFET gate	28 nm	Dark field
M2	Bottom electrode	100 nm	Dark field
M3	Pillar pattern	60 nm	Dark field
M4	Top electrode	80 nm	Dark field
M5	Disk pattern (weight-encoded)	80 nm	Clear field
M6	Contact vias	50 nm	Dark field
M7	Cu interconnect	1000 nm	Dark field
M8	Bond pad opening	50 μm	Clear field
M9	Passivation via	50 nm	Dark field
M10	Scribe line test structures	—	—

Total masks: 10. Mask M5 is generated from the weight data file described in Section "Weight Data Interface."

24 Metal Stack

The device uses a 10-layer copper metal stack.

Layer	Thickness	Pitch	Sheet R	Function
M1	100 nm	100 nm	0.20 Ω/sq	Local interconnect
M2–M5	200 nm each	200 nm	0.10 Ω/sq	Signal routing
M6–M7	400 nm each	400 nm	0.05 Ω/sq	Signal routing, clock
M8	1000 nm	1 μm	0.02 Ω/sq	Power distribution
M9	10 μm	10 μm	0.002 Ω/sq	Power distribution (thick)
M10	10 μm	10 μm	0.002 Ω/sq	Power distribution (thick)

Dielectric stack: standard low-κ (SiOCH) inter-metal dielectric.

25 Thermal Budget

Step	Temperature	Limit
FinFET	∼1000°C	$CoSi_2$ stable to 700°C
$CoSi_2$ formation	∼450°C	
Bottom electrode	300°C	TiN stable to 700°C
RTA anneal	450°C	HZO phase 400–600°C
Post-etch anneal	400°C	
Final anneal	400°C	Cu stable to 400°C

All temperatures are within limits.

26 Yield Analysis

26.1 Defect Density

Defect Type	Density
HZO pinholes	<0.1 cm^{-2}
TiN pillar defects	<0.01 cm^{-2}
Disk etch defects	<0.1 cm^{-2}
Passivation defects	<1 cm^{-2}
Particle contamination	<1 cm^{-2}
Total (mature)	**0.5 cm^{-2}**

26.2 Yield

For a 71 mm^2 die:

$$Y = e^{-0.5 \times 0.71} = 70.1\% \tag{4}$$

Good dies per 300 mm wafer: 670.

27 Cost Analysis

Parameter	Value
Wafer cost	$2,000
Good dies per wafer	670
Die cost	$2.99
Package cost	$5.00
Test cost	$1.12
Total chip cost	$9.11

Part V
Circuit Design

28 Supply and Reference

Parameter	Value
Logic supply	1.0 V
Write supply (programming only)	±3.2 V
Reference clock input	100 MHz, LVCMOS
Reference clock amplitude	1.0 V
Reference clock jitter tolerance	<10 ps RMS
Bandgap reference	1.2 V ± 1%

The write supply is only used during programming at fabrication. It is not required at runtime.

29 I/O Voltage Levels

Interface	Supply	Signal swing	Standard
Core logic	1.0 V	0–1.0 V	Core CMOS
PCIe 5.0 SerDes	0.9 V	Differential	PCIe 5.0
JTAG	1.0 V	0–1.0 V	LVCMOS
Reference clock	1.0 V	0–1.0 V	LVCMOS
Write/erase (test only)	±3.2 V	Analog	Custom

30 ESD Protection

Standard foundry I/O ESD protection is used on all pads.

Parameter	Value
HBM rating	2 kV
CDM rating	500 V
Implementation	Standard foundry I/O cells

31 Dual Word Line Drivers

Each input row uses two drivers: a non-inverting buffer and an inverting buffer.

31.1 Non-Inverting Buffer

Parameter	Value
Type	Class AB push-pull
Width (NMOS)	50 μm
Width (PMOS)	100 μm
Length	28 nm
On-resistance	1 Ω
Supply voltage	1.0 V
Signal range	0–50 mV
Charging time	2.25 ps
Number	1024 per layer per polarity

31.2 Inverting Buffer

Parameter	Value
Type	Class AB push-pull with inversion
Width (NMOS)	50 μm
Width (PMOS)	100 μm
Length	28 nm
On-resistance	1 Ω
Supply voltage	1.0 V
Signal range	−50 to 0 mV
Charging time	2.25 ps
Number	1024 per layer per polarity

31.3 Word Line Capacitance

$$C_{\text{wordline}} = 1024 \times 0.1 \text{ fF} + 20 \text{ fF} = 122.4 \text{ fF} \tag{5}$$

32 Pre-Charge Circuit

Each bit line is pre-charged to 0 V before each compute operation.

Parameter	Value
Type	Charge recycling
Pre-charge transistor width	4 μm
Pre-charge transistor length	28 nm
On-resistance	25 Ω
Pre-charge time	16.6 ps
Charge recycling fraction	50%
Storage capacitor	5 fF (standard MOS capacitor)

Charge recycling operation: The bit line charge is transferred to a storage capacitor instead of being dissipated to ground. The stored charge is used to pre-charge the next bit line. This reduces the pre-charge energy by 50%.

33 Analog Rectifier and Buffer Chain

The activation function is implemented as a current-mirror rectifier followed by a buffer chain.

Parameter	Value
Rectifier type	PMOS current mirror
Rectifier width	10 μm
Rectifier length	28 nm
Bias current	0.514 μA
Bias circuit	Diode-connected NMOS reference
Buffer stage 1	Drives 1 fF, width 10 μm
Buffer stage 2	Drives 10 fF, width 50 μm
Buffer stage 3	Drives word line (122.4 fF), width 100 μm
Supply voltage	1.0 V

34 Sample-and-Hold

Parameter	Value
Type	Switched capacitor
Sampling switch width	10 μm
Sampling switch length	28 nm
Sampling switch on-resistance	5 Ω
Hold capacitance	10 fF
Sampling time	1 ps
Droop rate	$<$1 mV/ms
Number	1024 per layer per network

35 Comparator

Parameter	Value
Type	Current-mode comparator with auto-zeroing
Topology	Double-tail latch with pre-amplification
Decision time	71.4 ps
Supply voltage	1.0 V
Power per comparator	2.347 μW
Auto-zero capacitor	100 fF
Reference current	1.0 μA
Offset (after auto-zero)	$<$10 μV
Number	1024 per network

36 Control Logic

Parameter	Value
Type	Finite state machine
Clock domain	Single, 500 MHz
Reset	Active-low, asynchronous
Register map	32 control registers, 32 status registers
Pipeline control	Layer pipelining, timing, calibration
Power	20 mW

37 ECC and Bad Block Management

Parameter	Value
Code type	SEC (Hamming)
Data word	64 bits
Check bits	7 bits
Code rate	0.90
Correctable errors	1 per 64 bits
Power	10 mW
Bad block table location	Block 0 (reserved)
Bad block table size	10% of total block count

Bad blocks are identified at test time and skipped during operation. The bad block table is stored in block 0 of the integrated memory.

38 Complete Power Summary

Component	Power
Comparators	336 mW
Analog rectifier + buffers	111 mW
Control	20 mW
I/O	50 mW
Clock	10 mW
ECC	10 mW
Pre-charge	1.81 mW
Word line drivers	2.04 mW
Scene decompressor	2 mW
Frame encoder	10 mW
Leakage	6 mW
Total	**560 mW**

39 Design for Test

- JTAG test access port.
- Built-in self-test (BIST) with neural network test vectors.
- Redundancy: 2 spare rows and 2 spare columns per sub-array.
- Laser fuse programming for repair.

Part VI
Operating Principles

40 Neural Network Weights

The neural shader weights are fixed at fabrication. They are determined offline and supplied to the foundry as a mask pattern file. The file format is given in Section "Weight Data Interface."

The weights are 1-bit at the baseline configuration. They are encoded as the presence or absence of a conductive region in each ferroelectric cell.

41 Programming at Fabrication

The neural network weights are programmed at fabrication during production test. The programming occurs only once. The weights cannot be changed in the field.

41.1 Write-Verify Voltage Levels

Attempt	Field	Voltage	Cumulative
1	1.5 MV/cm	2.2245 V	0.500
2	1.7 MV/cm	2.5211 V	0.875
3	1.9 MV/cm	2.8177 V	0.984
4	2.1 MV/cm	3.1143 V	0.999

41.2 Write Pulse Shape

Parameter	Value
Rise time	300 ps
Hold time	10 ns
Fall time	300 ps
Verify time	2 ns

41.3 Programming Time

With 100 cells programmed in parallel:

$$t_{\text{program}} = \frac{10{,}485{,}760 \times 22.2 \text{ ns}}{100} = 2.33 \text{ ms} \tag{6}$$

42 Compute Operation

42.1 Procedure

1. Pre-charge bit lines to 0 V (16.6 ps).
2. Apply differential input voltages to dual word lines.
3. Bit lines charge through cell conductances.
4. Sample-and-hold captures bit line voltage (1 ps).
5. Analog rectifier converts the output.
6. Buffer chain drives the next layer's word lines.

42.2 Compute Time

$$t_{\text{compute}} = \max(t_{\text{pre}}, t_{\text{WL}} + t_{\text{settle}} + t_{\text{charge}}, t_{\text{comp}}) \tag{7}$$

At the baseline configuration:

$$t_{\text{compute}} = \max(16.6, 3.45, 71.4) = 71.4 \text{ ps} \tag{8}$$

The comparator time dominates.

42.3 Pixel Rate

$$\text{Pixel rate} = \frac{N_{\text{active}}}{t_{\text{compute}}} = \frac{14}{71.4 \times 10^{-12}} = 196 \text{ Gp/s} \tag{9}$$

43 External Interface

43.1 Input

Compressed scene data is received over PCIe 5.0 at 64 GB/s. The scene graph decompressor expands the data into pixel attributes.

43.2 Output

Rendered frames are compressed by the frame encoder and transmitted over PCIe 5.0 at 64 GB/s.

43.3 External Pixel Rate

$$\text{Pixel rate}_{\text{external}} = 176.4 \text{ Gp/s} \tag{10}$$

44 Power Sequencing

Step	Timing
Apply 0.9 V PCIe SerDes supply	$t = 0$
Apply 1.0 V core supply	$t + 1$ ms
Apply 1.0 V I/O supply	$t + 2$ ms
Release reset	$t + 10$ ms
Load calibration	t+10 ms to t+10.1 ms
Begin operation	$t + 10.1$ ms

Reverse order for power-down.

45 Power-On Sequence

1. Apply 1.0 V logic supply.
2. Wait for power-on reset (1 μs).
3. Load calibration from on-chip memory.
4. Begin operation.

Total boot time: <10 μs.

46 Power-Down Sequence

1. Complete pending operations.
2. Remove logic supply.

The neural network weights are non-volatile. Power-off is instant. No state loss.

47 Error Handling

47.1 Correctable Errors

Correct the error using ECC, log the event, and continue operation.

47.2 Uncorrectable Errors

Retry the operation. If retry fails, mark the affected region as degraded.

Part VII
Performance Specifications

48 Summary of Baseline Performance

Parameter	Value
Internal pixel rate	196 Gp/s
External pixel rate	176.4 Gp/s
MAC throughput (1-bit)	2.06×10^{18} MAC/s
Compute time per layer	71.4 ps
Active networks	14
Power	560 mW
Efficiency	350 Gp/J
Die area	71 mm^2
Chip cost	\$9.11
Pipeline depth	10 layers
Logic supply	1.0 V
Operating temperature	0–85°C

49 Comparison with NVIDIA RTX 4090

Parameter	RTX 4090	This Device
Internal pixel rate	450 Gp/s	196 Gp/s
External pixel rate (PCIe 5.0)	—	176.4 Gp/s
External display output	2.5 Gp/s	176.4 Gp/s
Power	450 W	560 mW
Efficiency	1.0 Gp/J	350 Gp/J
Die area	608 mm^2	71 mm^2
Chip cost	\$30,000	\$9.11
Programmability	Full	None
Ray tracing	Yes	No

50 Comparison with AMD Radeon RX 7900 XTX

Parameter	RX 7900	This Device
Internal pixel rate	330 Gp/s	196 Gp/s
Power	355 W	560 mW
Efficiency	0.93 Gp/J	350 Gp/J
Die area	529 mm^2	71 mm^2
Chip cost	\$999	\$9.11

51 Honest Assessment

51.1 Strengths

- External pixel rate is 70.6× a flagship GPU's display output.
- Power is 804× lower than a flagship GPU.
- Efficiency is 350× better than a flagship GPU.
- Die area is 8.6× smaller than a flagship GPU.
- Cost is orders of magnitude lower.
- Non-volatile weights with instant boot.
- Multi-display capability: up to 353 4K displays at 60 Hz.

51.2 Weaknesses

- Internal pixel rate is 0.44× a flagship GPU.
- No programmability: the neural network weights are fixed at fabrication.
- No ray tracing.
- 1-bit weight precision.

Part VIII
Verification and Qualification

52 Overview

The verification plan follows the three-gate structure:

1. **Gate 1:** The ferroelectric phase is confirmed immediately after annealing.
2. **Gate 2:** The cell switches correctly at the single-cell level.
3. **Gate 3:** The device works at scale in an array.

53 Gate 1: Ferroelectric Phase Confirmation

53.1 When to Measure

After Step 10 (RTA Annealing).

53.2 What to Measure

A P-V hysteresis loop on a TiN/HZO/TiN capacitor.

53.3 Test Structure

A 50 μm × 50 μm capacitor fabricated on the same wafer, in the scribe line or a dedicated test area.

53.4 Equipment

Radiant Precision LC ferroelectric tester, or equivalent.

53.5 Procedure

1. Contact the top and bottom electrodes with probe tips.
2. Apply a voltage sweep from -5 V to $+5$ V and back.
3. Measure the polarization as a function of voltage.
4. Plot the P-V loop.

53.6 Pass Criteria

Measurement	Target	Pass
Hysteresis loop	Open loop	Clearly visible
Remanent polarization	>10 μC/cm^2	Above threshold
Coercive field	1–2 MV/cm	Within range
Film thickness	14.83 nm $\pm$ 1%	Ellipsometry

54 Gate 2: Single Cell Verification

54.1 Procedure

1. Probe the top and bottom electrodes.
2. Apply the write voltage (2.2245 V) for 10 ns.
3. Apply the read voltage (9.886 mV) and measure the current.
4. Apply the erase voltage for 10 ns.
5. Apply the read voltage and measure the current.
6. Repeat 10 times.

54.2 Pass Criteria

Measurement	Target	Pass
$n = 1$ read current	2.347 μA	Within 50%
$n = 0$ read current	<1 pA	Below threshold
ON/OFF ratio	$>10^9$	Above threshold
Reproducibility	10/10 cycles	All succeed

55 Gate 3: Small Array Verification

55.1 Procedure

1. Write all 4,096 cells in a 64$\times$64 test array to $n = 1$.
2. Read all cells and measure current.
3. Write all cells to $n = 0$.

4. Read all cells and measure current.

5. Calculate yield and cell-to-cell variation.

55.2 Pass Criteria

Measurement	Target	Pass
Array yield	>80%	Above threshold
Cell variation	<20% (1σ)	Below threshold
Write/read success	100%	No systematic failures

56 Reliability Testing

Test	Specification	Sample size	Acceptance
HTOL	1000h at 125°C	77 units	0 failures
HTSL	1000h at 150°C	45 units	0 failures
Temperature cycling	500 cycles, −40 to 125°C	77 units	0 failures
HAST	96h at 130°C/85% RH	77 units	0 failures
ESD (HBM)	2 kV	3 units	0 failures
ESD (CDM)	500 V	3 units	0 failures
Latchup	100 mA, 85°C	6 units	0 failures
Endurance	10^9 cycles at 85°C	30 units	0 failures
Retention	1000h at 85°C	30 units	0 failures

57 Burn-In

Parameter	Value
Temperature	125°C
Voltage	1.1× nominal
Duration	48 hours
Pattern	Neural network test vector + weight verification

Part IX
Failure Modes and Mitigations

58 Material Failure Modes

Mode	Cause	Severity	Mitigation
HZO not ferroelectric	Wrong anneal temperature	HIGH	Increase anneal to 500°C
HZO phase degradation	Grain growth	MEDIUM	Control RTA profile
Pinholes in HZO	Contamination	HIGH	Cleanroom, ALD QC

Mode	Cause	Severity	Mitigation
Film thickness error	ALD cycle count error	HIGH	Ellipsometry verification
Moisture absorption	Delay before anneal	HIGH	Anneal within 1 hour
Film thickness variation	ALD non-uniformity	MEDIUM	Ellipsometry at 9 sites

59 Device Failure Modes

Mode	Cause	Severity	Mitigation
Pillar over-etch	No etch stop	HIGH	Al_2O_3 etch stop layer
Disk over-etch	Poor selectivity	HIGH	Endpoint detection
Alignment error	Lithography drift	MEDIUM	± 2 nm overlay tolerance
Vortex not forming	Annular width insufficient	HIGH	Verify annular width $\geq 1.5\times$ core diameter

60 Circuit Failure Modes

Mode	Cause	Severity	Mitigation
Comparator offset	Transistor mismatch	MEDIUM	Auto-zeroing
Rectifier failure	Bias wrong	HIGH	Verify 0.514 μA bias
Pre-charge incomplete	Transistor too narrow	MEDIUM	4 μm width
Charge recycling failure	Storage capacitor leak	LOW	Accept reduced efficiency
Clock jitter	PLL noise	LOW	<1 ps RMS

61 System Failure Modes

Mode	Cause	Severity	Mitigation
Cell fatigue	Endurance exceeded	MEDIUM	Write-once (no wear)
Weight drift	Retention failure	MEDIUM	Periodic recalibration

Mode	Cause	Severity	Mitigation
Thermal runaway	Cooling failure	HIGH	Thermal shutdown
Read disturb	Read voltage too high	HIGH	Verify margin
Package warpage	Stress imbalance	MEDIUM	Stress compensation
PCIe transfer failure	Link error	MEDIUM	Retry protocol

Part X
Packaging

62 Package Specification

Parameter	Value
Package type	BGA
Ball count	1024
Ball pitch	1.0 mm
Package size	40 mm × 40 mm
Package height	1.5 mm
Thermal resistance	15°C/W
Die attach	Epoxy, 25 μm
Die bumps	Copper pillar, 50 μm pitch
Substrate layers	8 (4 thick copper for power)

63 Ball Assignment

Function	Ball count
Power (1.0 V logic)	256
Ground	256
PCIe 5.0 lane pairs	64 (32 TX + 32 RX)
JTAG	5
Reference clock	2
Reset	1
Test mode	4
Spare	436
Total	**1024**

64 Pad Ring Layout

The pad ring uses the foundry's standard pad ring generator.

Parameter	Value
Pad ring style	Staggered
Pad pitch	100 μm
Pad opening	50 μm $\times$ 50 μm
Pad type	Standard foundry I/O cells
Power pads	Distributed around perimeter
Signal pads	Ordered by function

The pad ring is generated from the ball assignment table using the foundry's standard pad ring tool. A reference pad ring script is available in the repository.

65 Floorplan Guidance

The device floorplan is organized into functional blocks.

Block	Area	Location
Neural shader crossbar (10 layers)	40 mm^2	Center
Comparators	8 mm^2	Adjacent to crossbar
Drivers (word lines)	4 mm^2	Adjacent to crossbar
Analog rectifier + buffers	5 mm^2	Between layers
Control logic	2 mm^2	Corner
ECC + bad block	1 mm^2	Corner
Scene decompressor	2 mm^2	Edge (PCIe side)
Frame encoder	4 mm^2	Edge (PCIe side)
PCIe SerDes	2 mm^2	Edge (PCIe side)
PLL + clock distribution	1 mm^2	Center
Pad ring	2 mm^2	Perimeter
Total	**71 mm^2**	

The floorplan places the neural shader crossbar in the center, the comparators and drivers immediately adjacent, and the PCIe interface at one edge.

66 Power Delivery Network

Parameter	Value
On-die decoupling	100 nF (distributed)
Package decoupling	10 μF (4 $\times$ 2.5 μF)
Board decoupling	100 μF
Target impedance	$<$10 mΩ from DC to 100 MHz
IR drop target	$<$5% of 1.0 V

67 Thermal Analysis

Parameter	Value	Notes
Power (typical)	560 mW	71 mm^2 die
Power (maximum)	700 mW	Worst-case workload
Thermal resistance	15°C/W	Package
Temperature rise	8.4°C	At typical power
Temperature rise	10.5°C	At maximum power
Junction temperature	33.4°C	At 25°C ambient, typical
Junction temperature	35.5°C	At 25°C ambient, maximum

All operating temperatures remain well within the 0–85°C range. No active cooling is required.

Part XI
Advanced Optimization Available

ADVANCED OPTIMIZATION AVAILABLE

The design specified in this document is the baseline implementation of the ferroelectric vortex GPU platform.
Further optimization is possible.
The advanced optimization provides:

- **Internal pixel rate:** 1,090 Gp/s (5.56× the baseline)
- **External pixel rate:** 176.4 Gp/s (unchanged)
- **MAC throughput:** 1.14×10^{19} MAC/s (5.56× the baseline)
- **Power:** 726 mW
- **Efficiency:** 1,502 Gp/J (4.29× the baseline)
- **Die area:** 71 mm^2 (unchanged)
- **Cost:** $9.12

The complete specifications for the advanced optimization are available from the author.
Contact: M.V.Hristov@proton.me
The advanced optimization is not included in this document.

Part XII
Appendix A: Complete Numerical Tables

A Fundamental Engineering Parameters

Parameter	Value
Cell size	120 nm × 120 nm
Disk diameter	80 nm
Pillar diameter	60 nm
Pillar height	1.96 nm
Film thickness	14.83 nm
Annular width	10 nm
Conductive region effective radius	2.31 nm
Resistance ($n = 1$)	4,212 Ω
Capacitance	97.8 aF
Settling time	0.412 ps
Read voltage	9.886 mV
Read current	2.347 μA
Write voltage (attempt 1)	2.2245 V
Write voltage (attempt 4)	3.1143 V
Internal pixel rate	196 Gp/s
External pixel rate	176.4 Gp/s
MAC throughput	2.06×10^{18} MAC/s
Compute time per layer	71.4 ps
Power (typical)	560 mW
Power (maximum)	700 mW
Efficiency	350 Gp/J
Die area	71 mm^2
Chip cost	\$9.11

B Material Specifications

Parameter	Value
Ferroelectric composition	$Hf_{0.5}Zr_{0.5}O_2$
Ferroelectric thickness	14.83 nm
Coercive field (25°C)	1.49 MV/cm
Coercive field (85°C)	1.18 MV/cm
Curie temperature	579 K
Remanent polarization	13.07 μC/cm^2
Dielectric constant	30
Resistivity	1.284×10^{-4} Ω·m
Bottom electrode	TiN, 50 nm
Top electrode	TiN, 50 nm
Pillar	TiN, 1.96 nm

Parameter	Value
Etch stop	Al_2O_3, 2 nm

C ALD Recipe Summary

Step	Precursor	Pulse time	Purge time
1	$HfCl_4$	0.1 s	3 s
2	H_2O	0.1 s	3 s
3	$ZrCl_4$	0.1 s	3 s
4	H_2O	0.1 s	3 s

Cycle ratio: 1 Hf : 1 Zr.
Supercycle growth: 0.18 nm.
Supercycles: 82.
Deposition time: 17 min.
Deposition temperature: 250°C.
Chamber pressure: 0.5–2 Torr.

D Circuit Specifications

Parameter	Value
Comparator decision time	71.4 ps
Comparator power	2.347 μW
Word line driver power	50 μW
Write driver voltage range	$\pm$3.2 V
Write driver transistor type	5 V I/O
Pre-charge time	16.6 ps
Pre-charge recycling fraction	50%
ECC code type	Hamming SEC
ECC data word	64 bits
ECC code rate	0.90
Scene decompressor power	2 mW
Frame encoder power	10 mW

E Fabrication Summary

Component	Steps	Masks	Notes
GPU die	18	10	Standard HZO process

Part XIII
Appendix B: Glossary

Term	Definition
Neural shader	A fixed neural network used to compute pixel colors instead of programmable shader code
Dual word line	Architecture for signed weights using two word lines per input
Ferroelectric cell	A memory cell that stores information in the state of a ferroelectric material
HZO	Hafnium zirconium oxide, $Hf_{0.5}Zr_{0.5}O_2$
ALD	Atomic layer deposition
RIE	Reactive ion etching
RTA	Rapid thermal annealing
ECC	Error correction code
BGA	Ball grid array
TIA	Transimpedance amplifier
DAC	Digital-to-analog converter
ADC	Analog-to-digital converter
MAC	Multiply-accumulate
Internal pixel rate	Rate at which the device processes pixels internally
External pixel rate	Rate at which rendered frames are transferred to the host
ReLU	Rectified linear unit, the activation function
HTOL	High temperature operating life test
HTSL	High temperature storage life test
HAST	Highly accelerated stress test
ESD	Electrostatic discharge
HBM	Human body model (ESD test)
CDM	Charged device model (ESD test)
BIST	Built-in self-test
DFT	Design for test
PCIe	Peripheral Component Interconnect Express
PDN	Power delivery network

Part XIV
Appendix C: Licensing

F Document License

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G Technology License

The technology described in this document is released as a **public, patent-free engineering specification**.

No patent is sought. No license is required to manufacture, sell, or use devices implementing the design described in this document.

H Repository Access

Complete specifications, updates, and supporting materials are available at:

- **Main Repository (Mega):** Mega
- **Proton Drive:** Proton Drive
- **Zenodo:** Zenodo
- **GitHub:** GitHub
- **OSF:** OSF
- **Academia.edu:** Academia.edu
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